

PAVEL SKRIPKIN

SENIOR SYSTEMS SOFTWARE ENGINEER / OS KERNEL ENGINEER

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C • Rust • OS Kernels • Memory Management • Low-Level Concurrency • Kernel Hardening • Performance

PROFILE

Senior systems software engineer with 6 years of experience building production OS kernels and low-level platform software. Focused on memory management, low-level concurrency, kernel hardening, and ARM64 (ARMv8); contributor to Linux, LLVM, and QEMU.

EXPERIENCE

Kaspersky Senior Systems Software Engineer / OS Kernel Engineer

Aug 2023 - Present

Moscow • Core-kernel and kernel-hardening teams in KasperskyOS.

- Designed and implemented user-space pagers, enabling paging and swap policies outside the kernel.
- Implemented inline kernel AddressSanitizer instrumentation, improving ASan-instrumented kernel performance by approximately 6x.
- Implemented kernel hardening mechanisms including shadow stacks, KCFI, and ARMv8 PAN.
- Implemented zero-page deduplication for zero-filled memory pages.
- Debugged scheduler-related and concurrency issues in the production kernel.
- Integrated static analysis workflows and contributed improvements to Clang Static Analyzer support.

Samsung Systems Software Engineer / OS Kernel Engineer

Nov 2021 - Aug 2023

Moscow • Proprietary ARM64 (ARMv8) OS kernel with Trusted Execution Environment (TEE) functionality.

- Implemented full Device Tree Blob (DTB) support and ARMv8 platform integration.
- Implemented kernel reader-writer locks for low-level synchronization.
- Implemented read-only page tables as a kernel-hardening mechanism.
- Developed TEE components and debugged low-level kernel and ARMv8 platform issues.

NTC Raduga Low-Level Software Engineer

Jun 2020 - Nov 2021

Moscow • Linux kernel drivers for FPGA-based hardware platforms.

- Developed Linux kernel drivers and user-space utilities for FPGA-based hardware; handled low-level integration and debugging.

TECHNICAL SKILLS

Programming Languages: C, Rust, C++

Kernel / OS: memory management, user-space paging, Linux kernel, device drivers

Concurrency: synchronization primitives, reader-writer locks, lock-free algorithms

Architecture / Platform: ARM64 (ARMv8), x86-64, DTB

Security / Hardening: TEE, KCFI, PAN, shadow stacks, AddressSanitizer

Performance: perf, AVX2, cache behavior, branch prediction

Analysis / Debugging: Clang Static Analyzer, low-level debugging

OPEN SOURCE CONTRIBUTIONS

Contributor to the [Linux kernel](#), [LLVM](#), and [QEMU](#).

Nominated for the Google Open Source Peer Bonus (2023) for Linux kernel contributions. [Announcement](#)

SELECTED TECHNICAL PROJECTS

Custom OS [GitHub](#)

Rust microkernel-style OS with a custom async scheduler/runtime, IDL bindings, asynchronous IPC, and ARMv8 support.

Lock-free buddy allocator [GitHub](#)

Lock-free buddy allocator based on an academic algorithm.

EDUCATION

National Research University Higher School of Economics, MIEM

BSc in Computer Security • 2018 - 2024

Thesis: CPU jitter-based true random number generator (TRNG).

LANGUAGES

Russian (native) • English (B2+)